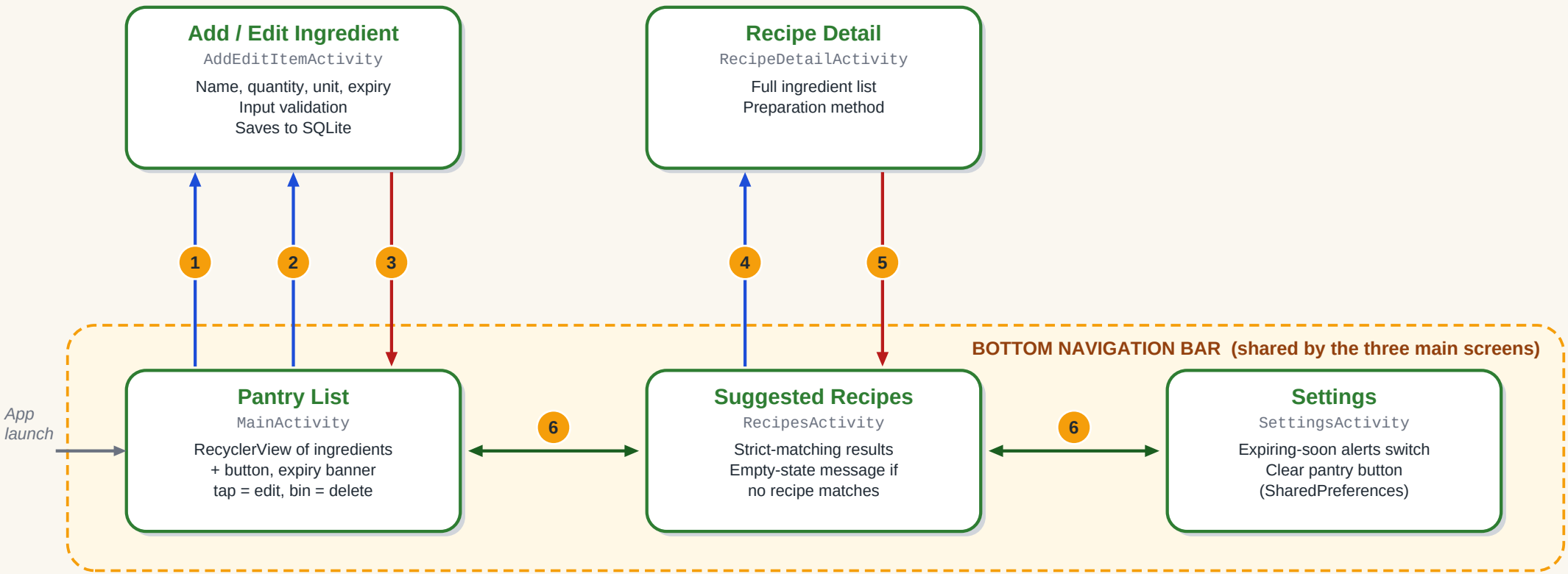
Screen Flow Diagram

How the five screens (Activities) connect. Numbered arrows are explained in the key below.



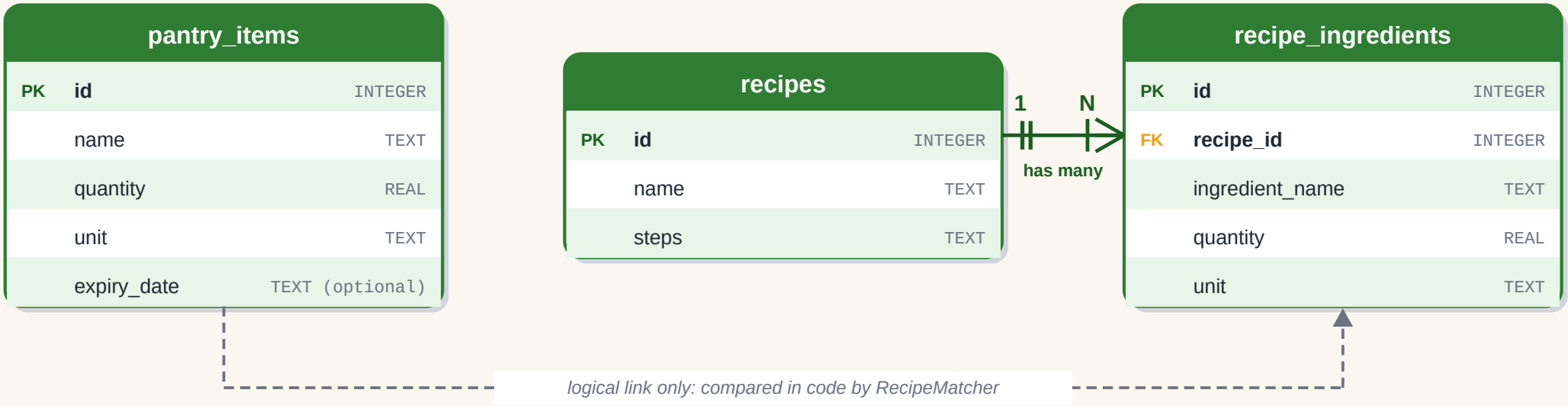
Key

1. Pantry List to Add/Edit: + button. Intent with no extras = ADD mode.
2. Pantry List to Add/Edit: tap an item. Intent + extra "item_id" = EDIT mode (item loaded from SQLite).
3. Add/Edit to Pantry List: Save (insert or update, then finish) or Back arrow.
4. Suggested Recipes to Recipe Detail: tap a recipe. Intent + extra "recipe_id".
5. Recipe Detail to Suggested Recipes: Back arrow.
6. Tabs on the bottom navigation bar open each main screen with an Intent (UiUtils.setupNav).

Opens a screen (Intent) — Returns / finishes — Bottom-nav tab switch (both ways)

Data Model / ER Diagram

SQLite database smart_pantry.db (SQLiteOpenHelper) - three tables



Notes

PK = primary key (auto-incrementing). FK = foreign key.

recipes (1) to recipe_ingredients (N): recipe_ingredients.recipe_id references recipes.id, with ON DELETE CASCADE.

pantry_items has no foreign key. RecipeMatcher compares pantry rows with recipe_ingredients rows in Java code:

names are normalised (tomatoes = tomato, spaghetti = pasta), quantities are converted to base units (g, ml, piece),

and a recipe is suggested only if EVERY ingredient is present in at least the required quantity.

Recipes and ingredients are seeded once in DatabaseHelper.onCreate (18 recipes); pantry_items supports full CRUD.